MODULE 2_APA: INTRODUCTION TO CUSTOMER DATA ANALYTICS

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BIM, Trichy

CDP DATA PLATFORMS

Org. Data Platform

Need for a Data Platform

01

Various **Enterprise Data Sources** including Core Business Operations generate useful **structured data** (Customer, Product, Transactions, etc)

02

To unify and bring them to one place conventionally **Data Warehouses** were used

03

Other **external and Web sources** generate huge **unstructured data** (emails, Website, Partners, Social media – Big data)

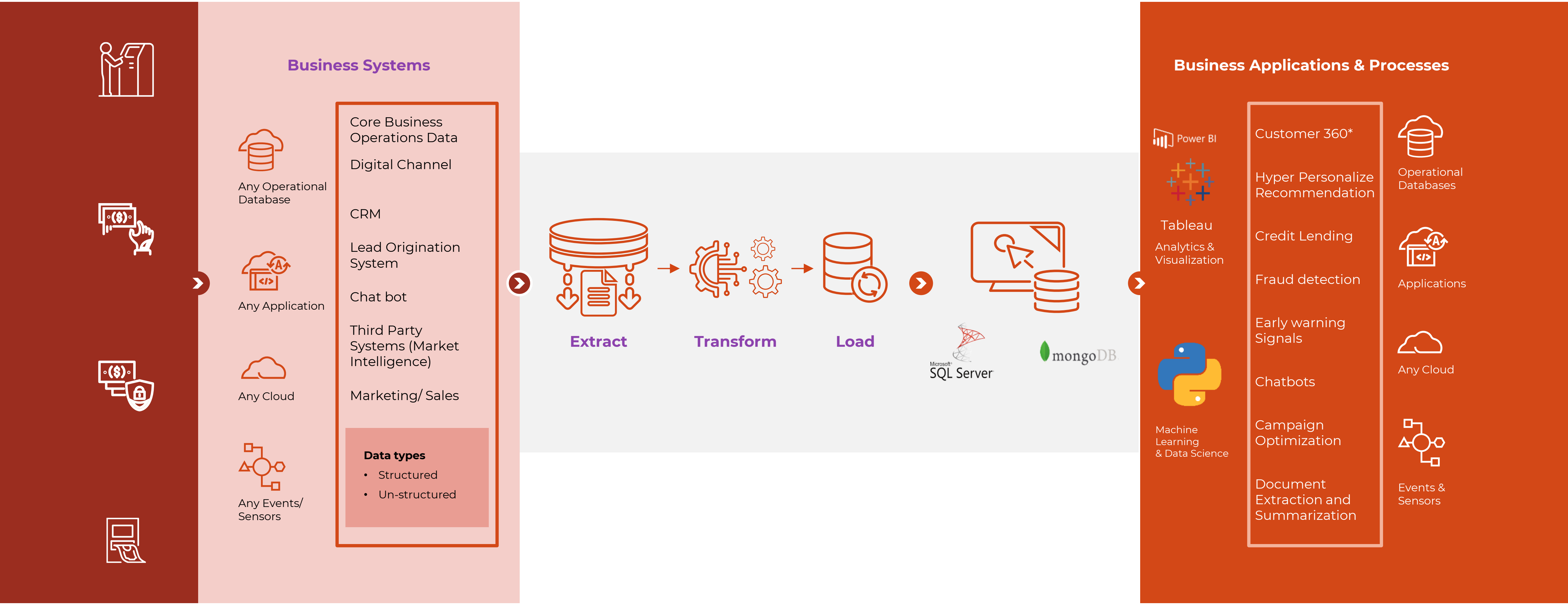
04

To find useful data among **data streams** and filter and store them in **raw form** generally **Data Lakes** are used

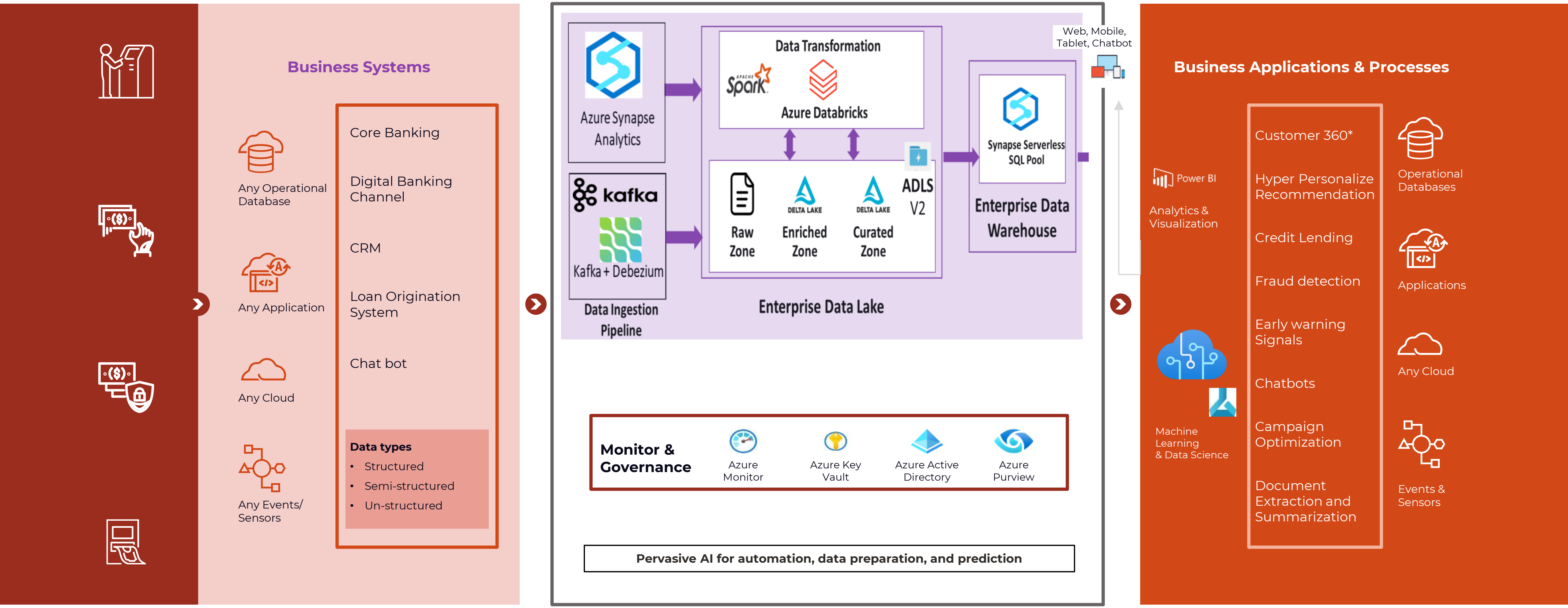
05

For ensuring Analytics to **assess trends a**nd AI/ML models to predict **Product and Spend** etc Data Lakes data are funnelled as useful datasets

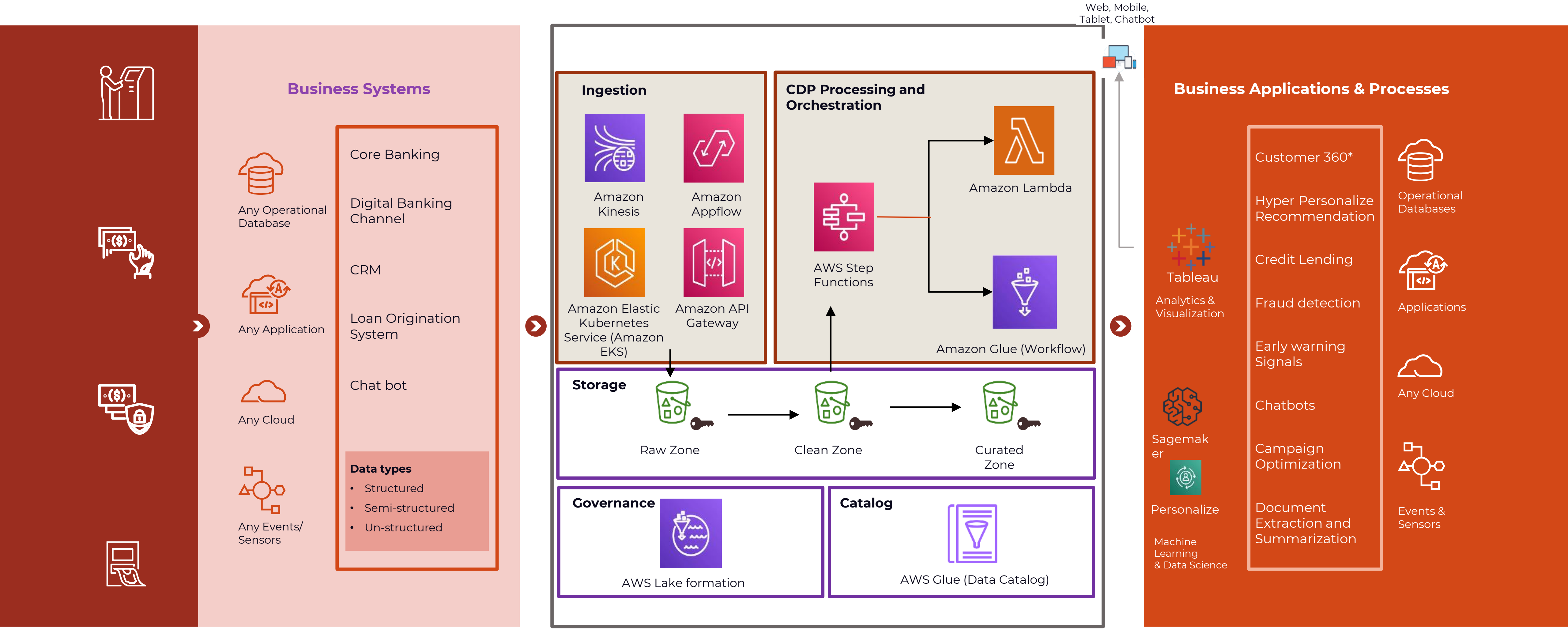
On-Prem - Data platform



Cloud Data platform on Azure Data Lake



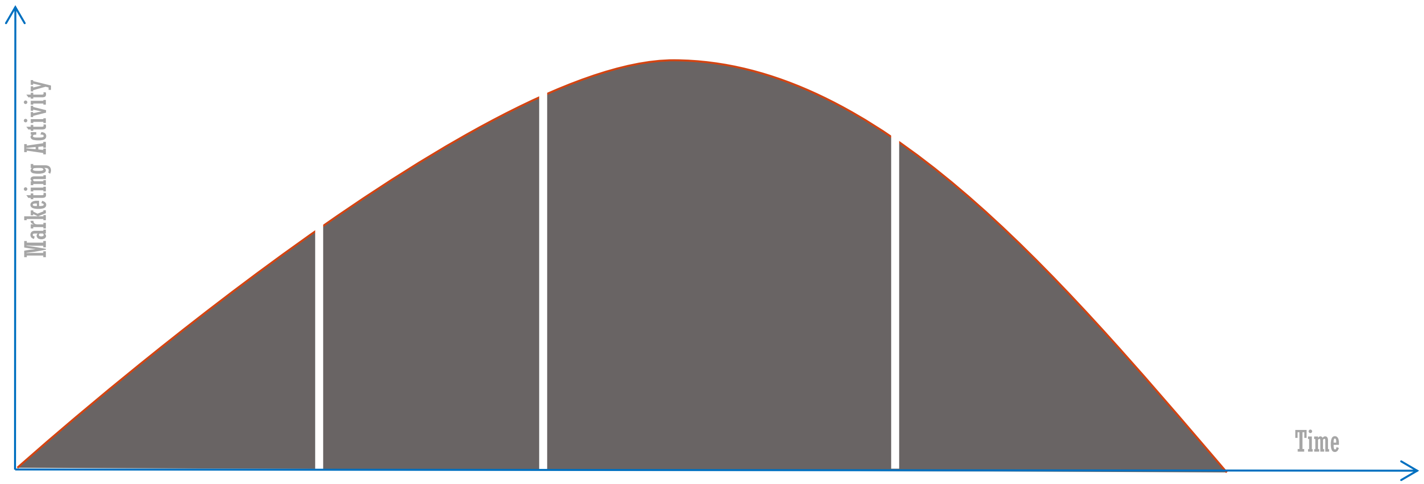
Cloud Data platform on AWS Data Lake



INTRODUCTION TO CUSTOMER DATA ANALYTICS - CONCEPT

Sridevi Tandley

ADVANCED PREDICTIVE ANALYTICS CAN HELP ADDRESS A VARIETY OF BUSINESS CHALLENGES THAT ORGANIZATIONS FACE ACROSS THE CUSTOMER LIFECYCLE …



Strategy:	Acquire	Engage	Develop	Retain	Win-back
Capabilities:	Targeting Analysis	Analysis Test Design Reporting	Analysis Test Design Reporting	Optimization Test Design Reporting	Targeting Analysis
Plays:	• Which Customers will buy BFSI products & services • Who will be buying a 'repeat product & services'	• What will these customers buy next from us?	• Can I spend my Marketing money better? • What kind of bundles can I create to sell them?	• Which Marketing campaigns works the best for these customers?	• What should I sell to win these customers back • Who will buy, and who will not buy

How to leverage customer data across the customer lifecycle to meet business objectives?

PREDICTIVE ANALYTICS LEVERAGES CUSTOMER DATA TO ANSWER KEY QUESTIONS OF ORGANIZATION

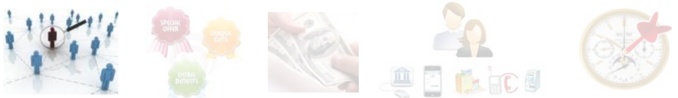
WHOM TO TARGET?

WHAT TO OFFER?

WHAT IS THE PRICE?

WHICH IS THE BEST CHANNEL?

WHEN TO TARGET?



1 WHOM TO TARGET – HELPS IDENTIFY THE RIGHT SET OF CUSTOMERS TO TARGET

Based on the business objective, we can analyze customer data to develop Analytical models, which will help identify the right target base

Objective: Rank order customers based on likelihood to respond to a campaign

Analyze customer data - Demographic, Transaction data, Product holding data, Credit Bureau data, Past campaigns & Customer interaction data (with Call center, etc.)

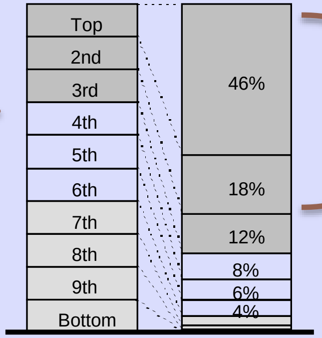
Segment customer base depending on business objective

Identify differentiating characteristics amongst the various segments through statistical techniques

Illustrative

Increase spend per card - Whom to target?

- Credit Card Customer Base
  - Low Value (LV)
    - High propensity to move to MV/HV
    - Low propensity
  - Medium Value (MV)
  - High Value (HV)



- By targeting 30% of Population, we can capture 76% of potential responders
- Typical characteristics of High propensity customers – young, have increasing utilization in 6 months, higher number of credit cards transactions at online shopping, departmental stores & high outstanding balance on personal loan

Target all HV & MV + Top 30% (by propensity) of LV

ENHANCED TARGETING EFFICIENCY FOR CREDIT CARDS – LOWER COST PER ACQUISITION (CPA) GLOBAL BANK

1

Objective: Identify right Acquisition strategy for Credit Cards division, such that CPA is lowest

NA has larger presence of Offline Mail compared to Digital E-Mail. So the initial CPA was way higher.

Target Audience need to be identified rightly based on the Card Usage potential with lower Delinquency Rate

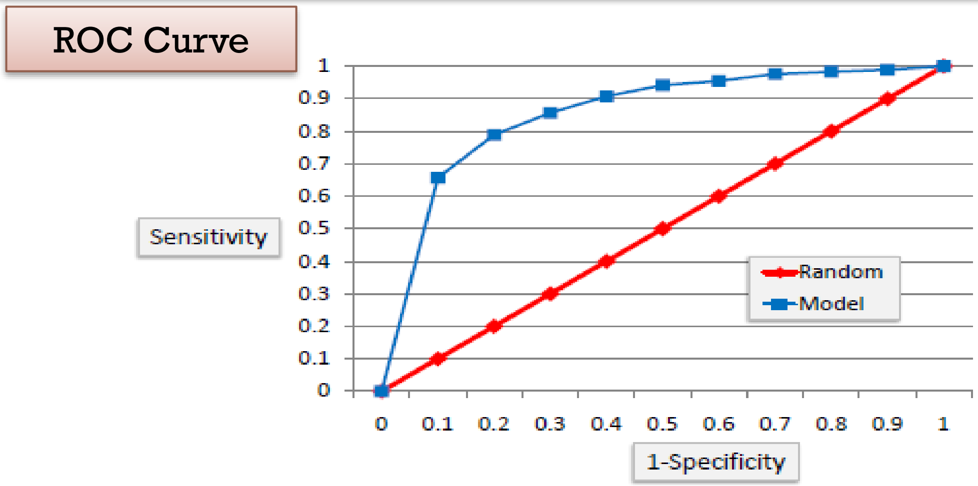
3

Approach: Develop a propensity to identify potential customers for lending products

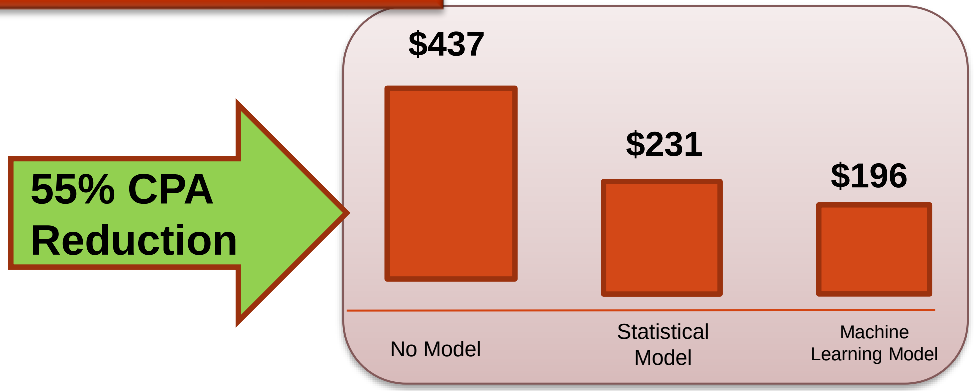
- ✓ Sub-channel analysis carried out for New Card launch, New Card specific response & behavior models created
- ✓ Moved from rule based segment to ML technique (XGB, GBM, RF)
- ✓ New bureau features created leveraging Customers Balance Volatility for last 6 months using CNN model
- ✓ Track & measure responses to each campaign and use it to test the filters and develop propensity model

2

Delivery: Enhanced targeting efficiency – Lower CPA, more accounts with high propensity to revolve



Estimated Benefits



IMPROVE LOAN VOLUMES BY 22% THROUGH CROSS SELL

1

Objective: Identify pure funding customers who are eligible for a pre-approved Quick Cash loan based on their current level of engagement

82% of funding customers do not have loan products

Huge attrition in lending customer base in 2007 (over 5000 customers) has lead to a fall in lending volumes

2

Approach: Develop a propensity to identify potential customers for lending products

Arrive at most eligible customers by determining filters to be applied to overall CASA base

Run targeted campaigns

Track & measure responses to each campaign and use this to test the filters and develop propensity model

	Filters at Customer level	# CUSTOMERS
1.0	Complete CASA base as of Dec'07	56584
1.1	Currently Active customers	43598
1.2	Complete CASA base as of Dec'07 with no lending for last 6 months, but only funding	33897
2	Frequency - atleast 3 transactions a month for last 6 months	19203
3	Monetary value - less than 3 transactions > 50 Mn (each txn) in last 3 months	15378
4	Avg balance for last 6 months >= 10 Mn	4723
5	Months of relationship >= 12 months	4394
6	Average credit transaction for the last 6 months >= IDR 20 mio	2955
	Eligible Base as of Dec'07	**2955**

3

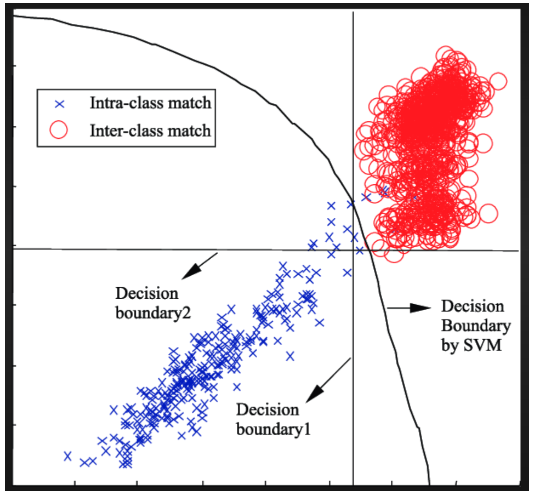
Result: Estimated Increase of 0.6% in annual lending Net Interest Income

- ✓ Expected conversion rate of 22% from a 3 month campaign
- ✓ This is assuming a plafond size of 3 times the average balance and 70% utilization

Hyper-personalisation is the biggest revenue driver of the Center of Excellence

Challenges

- Many campaigns based on traditional segmentation-based targeting; no personalisation
- Lack of targeted customer reach
- Plain vanilla offers for all customers within segment



Hyper-Personalised Decision Boundaries for SVM using RBF Kernel function

Solution

Created a data lake to collate historic evidence and live data feeds

Developed 40+ predictive models using **SciKi**t learning packages to predict baseline:

- **SimpleImputer** – to impute missing value,
- **OneHotEncoder & CategoricalEncoder** – to convert category into binary attributes
- **StandardScale**r for standardization
- **RandomizedSearchCV** to fine turn the hyperparameters
- Some Models API used are – **liner_model, kernel_ridge, SVM, tree, multiclass**

Developed deep learning model to estimate the possible stretch using TensorFlow

- **DNNClassifier** to create neuron layers
- **tf.layer.dense** create a fully connected neuron network
- **Cross Entropy** as cost function to penalized model estimate low probability

Developed deep learning model to estimate the possible stretch using **TensorFlow**

Created an infrastructure on GCP for

- to create data lake (historic behaviour plus live data feeds at customers level)
- to store model and targeting algorithm output (prioritized offer for every customers)
- Python code to create a list pull daily and upload on StrongMail http

Benefits

4%
Increase in registration rate

$1.6B
Total revenue; **33%** higher

+$440M
Incremental revenue; **23%** above target

+360,000
New members enrolled for program

ATTRITION AND RETENTION: APPROACH AND BENEFITS

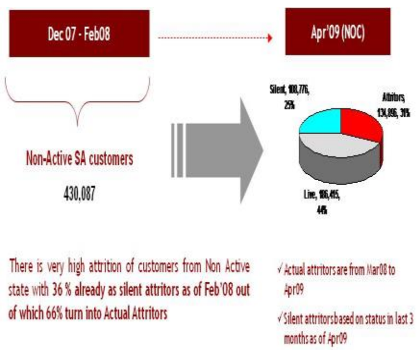
Business Challenge/ Need

To identify accounts that are likely to attrite in the immediate future so that these can be targeted through suitable retention campaigns

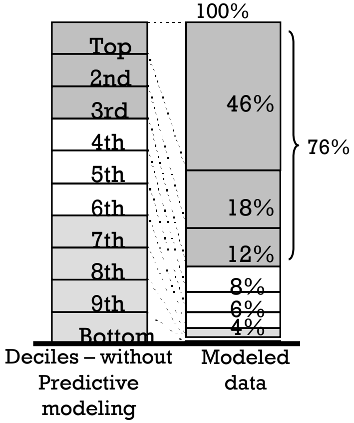
Our Approach

Illustrative

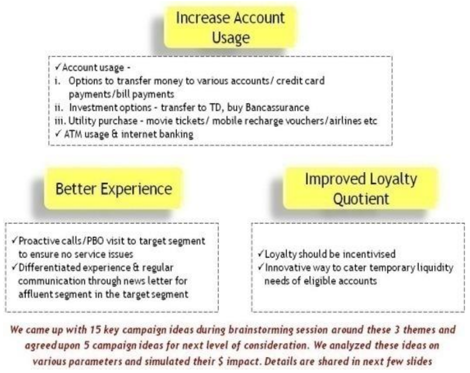
Identify attrition rate present in portfolio



Identify customers who are more likely to attrite in recent time



Identify central themes for retention programs



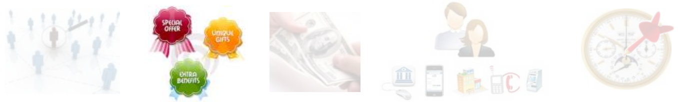
- Find the customer base you want to run retention campaigns on
- Identify key attributes that determine customers likely attrite
- Identify high-propensity customers for retention campaign based on behavior score
- Launch the targeted campaign on the high-propensity base

Benefits / Impact

✓ Reduce Churn / Attrition
✓ Improve Retention Rates
✓ Identify & address drivers for attrition proactively

Typical data needed

- Customer Demographic Data – static
- Product holding data - historical for last 6mths to 1 year
- Existing Customer banking transactional data
- Campaign results for last 6 months (optional)



2 WHAT TO OFFER - HELPS IDENTIFY RIGHT PRODUCT / SERVICE / CAMPAIGN FOR EACH CUSTOMER SEGMENT

To understand what best to offer to customers, we leverage all types of data (structured data and unstructured data – both internal and external)

Objective: To identify and offer most relevant offer to customers

Identify right technique based on context - Segmentation / Spend purchase analysis / Next best action / Sentiment analytics

Analyze Transaction / Product holding data with regards to customer profiles as well as Unstructured data (internal, social media and external) if available

Develop campaigns based on insights from the analytics models

Illustrative

Increase spend per card – What to offer?

A Based on same customer, past usage

Profile a customer's past Card transactions basis location, time / date / day of spend, merchant categories / subcategories, spend amount, etc and identify appropriate rules

Identify **right promo** for **each customer**; Eg. – LV, Likely to move to MV, used card at Japanese restaurants in past – therefore focus on lunch offers at Japanese restaurants with increased relevance based on customer location and timing of campaign

B Next best category - Based on similar customers, current usage

Analyze transactions of customers by merchant category

Customer	Transaction at merchant category
1	Apparel, Watch
2	Electronic App, Apparel, Mobile
3	Apparel, Footwear
4	Apparel, Footwear, Watch
5	Mobile, Watch

Generate co-occurrence matrix for 1/2/3 transactions

	Apparel	Mobile	Electronic App	Airlines	Footwear
Apparel	4	1	1	2	2
Mobile	1	2	1	1	0
Electronic App	1	1	1	0	0
Airlines	2	1	0	3	1
Footwear	2	0	0	1	2

Develop association rules

- If customer did recent transaction at **Apparel** then most likely he is interested in **Airlines**
- If Retail, **Apparel and Mobile** then **Jewellery**

IMPLEMENTED CROSS SELL MODEL FOR A MIDDLE EASTERN BANK…

1 **Objective:** To indentify potential customers within its banking portfolio to target for cross sell of insurance products

- The client was interested to know which products should be focused upon
- Client was also keen on us helping it with the implementation and tracking of the cross sell campaigns

2 **Approach:** We developed a hybrid matrix based on customer needs and mapped the offered products in each of the cells

- Tapped into our understanding of customer value benchmark elements :
  - relative income levels
  - debt burden ration
  - average credit amounts in last six months customer loyalty
  - local attributes of demographics
  - bank specific product holdings
- We implemented the recommendations through a test and control method before large scale implementation

Customer Age						
Young <= 35	A & H Term Plan	Basic CI A & H Term Plan Endowment	CI with ROP Whole of Life Endowment Savings ULIP	CI with ROP Whole of Life Education Retirement Savings & Inv ULIP	SP CI Plan Whole of life Retirement NC Plan	SP CI Plan Whole of Life Retirement Lump sum Inv
Middle Age 35 to 45	A & H Term Plan	Basic CI A & H Term Plan Endowment	CI with ROP Whole of Life Endowment Retirement Savings	Basic CI Whole of Life RTA Education Retirement Savings & Inv ULIP	SP CI Plan Whole of Life RTA Retirement Savings & Inv	SP CI Plan Jumbo Life Retirement Lump sum Inv
Pre - Retirement > 45	Stepped down CI A & H Term Plan	Stepped down CI A & H Term Plan Endowment Savings	Basic CI Term Plan Endowment Savings & Inv	Basic CI Whole of life Endowment Savings & Inv	SP CI Plan Whole of Life Lump sum Ins Savings Lump sum Inv	SP CI Plan Jumbo Life Retirement MT lump sum Inv PPLI Lifestyle Ins
Income of the customer (AED) / Average balance / PFA (AED)	<= 5K / <= 20K	5k to 12k / 20k to 50k	12k to 20k / 50k to 100k	20k to 35k / 100k to 200k	35k to 50k / 200k to 300k	> 50k / > 300k

Product preference matrix based on customer lifestage, demographics and needs

3 **Result:** Strategies for **Bancassurance** cross-sell were differently defined for different nationalities

- While customer needs and income defined the type of product to be offered to the customer, **'affordability'** played a big role in leading the customer to finally accept the offer
- We observed that the preferences and needs varied significantly with the **nationality** of the customer, since the bank has a significant mix of 5 different nationalities.

What to Offer - Case Study

STRATEGIC LOYALTY MEMBER SEGMENTATION SCHEME FOR A CUSTOMIZED TARGETING PROGRAM

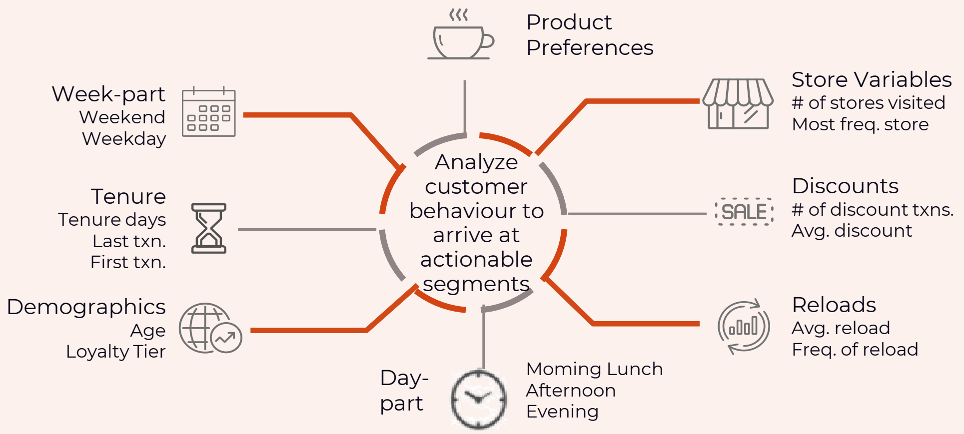
Business Needs

- Incorporate a futuristic transformation solution over the existing customer targeting strategy that was based on hard to customize business rules
- Integration of the solution with marketing initiatives for long term growth
- Improve campaign uptake and redemption rates in comparison to the existing mechanism

Solution

- Customers were segregated into different engagement tiers (macro-segments) based on the length of their association with the retailer
- Models were built using various internal and external sources of data to identify micro-segments within each
- These segments were used to devise better targeting strategies and offers leading to higher levels of engagement with the retailer

Amalgamate business priorities and customer insights through data mining to craft actionable macro-segments



Snapshot of few variables used for analysis

Estimated Benefits

$20M
incremental revenue through strategic targeting

52,000
Customers migrated from low to high engagement segments

8%
Higher uptake on campaigns based on segmentation

Recommender System – Quick Online Service

Business Objectives

Build recommender system for the mobile app of popular pizza company franchise.

It shall recommend/sort user preference for pizza, base type, bread type, toppings, sides, deserts through out purchase journey on the mobile app/website which will improve customer experience.

Approach

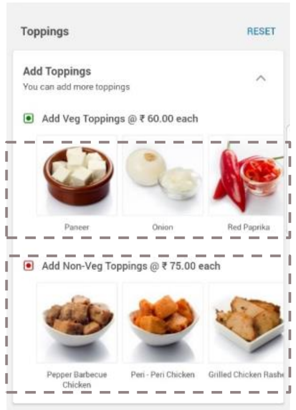
Data Pipelines/ Streaming
Build ETL pipelines to build/ manage batch and streaming data coming to the data management platform

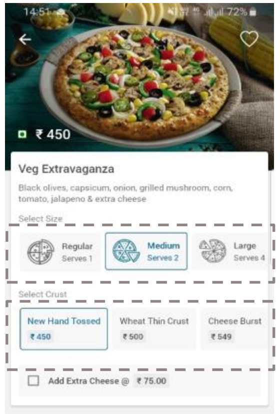
Define Objective Function
Define success objective across all the pages mobile app/ website

Top N Recommender System
Build separate recommender system across sub problem defined in earlier stages – Deep learning based MF models (xDeepFM)

Deployment
This includes API connection deployment with their product and our solution construct

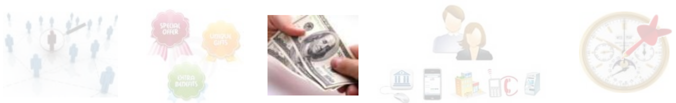
Solution Overview





Business Impact

- Hyper-personalized recommendation has improved user buying experience on app
- Bad-orders has gone down significantly
- Opened path for "30 to 20" campaign strategy in the near future



3 WHAT IS THE PRICE – HELPS OPTIMIZE THE PRICE POINT FOR THE OFFER

To understand price sensitivity of a customer based on proxies which can be understood from transactional and behavioral data

Objective: Identify price sensitivity or price consciousness of a customer

Analyze proxy transaction variables that showcase a customer's tendency to be price sensitive in his/her daily life

Develop a segmentation model to bucket customers into different levels of price sensitivities

Map sensitivity to customer profitability and develop a pricing strategy for each segment

Illustrative

Increase spend per card – What is the price?

Develop a library of proxies → …Apply these proxies to the customer base to identify price sensitive or conscious customers → …Design a delta pricing discount to be offered to appropriate customers (assuming a base price for each product/campaign applicable to all customers)

Examples of price sensitive customers:

- Someone who pays his entire credit card bill within 1-3 days of receiving the statement
- Someone who always calls up for getting annual fee waived off
- Someone who takes the shortest tenure loans as possible
- Someone who keeps changing his bank for his home loan or prepays the home loan …etc

PRICING SCENARIO PLANNING TOOL

Pricing - Case Study

Objectives

- CPG manufacturers would like to understand the impact of price changes in own brand / competition based on volume change across the various channels.

  They required insights on:
  - How sensitive is my volume to price changes?
  - How can I set a pricing range that protects my market share?
  - Is the optimal price range different across retailers?
  - Which brands' prices impact my brand and how?

Approach

- Log-linear regression model needs to be built to obtain own & cross price elasticities
- Price corridors are to be made to suggest a range within which own price index with respect to a competitor needs to lie in order to maintain the current market share
- Across channel comparisons are to be done to identify the channels, which are more price sensitive and which are less

3 Step Analysis Plan for pricing analysis

Individual model	Competitor aggregate model	Price corridors
Identify key competitors impacting client's volume share	Build a model with aggregated competitor presence	Identify price corridors against each of the competitors as well as against the aggregated competitor presence

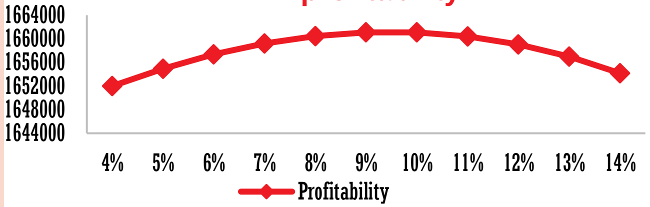
Analysis/ Action Taken

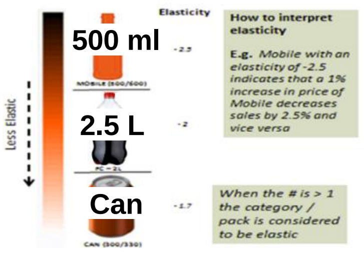
- Designed a Pricing Scenario Planning Tool, which creates various scenarios based on price elasticity and competitor price elasticity to understand their impact on brand's volume.

Sample Output

Volume = **f{**Own Price, Competitor's Price, Promotions, Seasonality**}**

Optimal discount point to maximize profitability

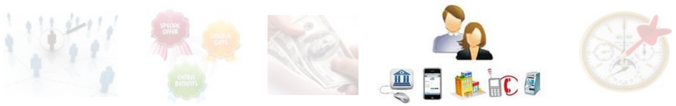




Impact

Identified low price elasticity, so pricing & promotion would not significantly grow category penetration

20% increase in volume by change in price



4 WHICH IS THE BEST CHANNEL – HELPS MAXIMIZE RESPONSE RATE BY IDENTIFYING THE RIGHT EXECUTION CHANNEL FOR A CAMPAIGN

Channel optimization provides insights into a customer's preference as well as the possibility of migrating him to lower/higher cost channels

Objective: To Identify the right execution channel for each customer

Analyze customer profitability, transactions performed across channels, cost of channels, etc.

Develop channel optimization model

Overlay model with capacity available on the channels / availability of updated contact details of customer

Illustrative

Increase spend per card – Which is the best channel?

Channel usage analysis at a customer level (# and types of transactions at each channel)

+

Calculate customer profitability (can use revenue as a proxy)

Customer ID	Customer Profitability	Segment
A1	448	HIGH
B1	337	Medium
C1	219	High

+

Map channels to customer

Customer	B1
Channels Used	Call Centre & Internet
Current Profitability	Medium
Channels preferred (based on What to Offer)	SMS, Online, Call centre

Offer campaign to Customer B1 through Internet banking channel

Channel Identification & Optimization

Helps maximize response rate by identifying right execution channel for a campaign (1/2)

Illustrative

Channel optimization provides insights into a customer's preference as well as the possibility of migrating him to lower/higher cost channels

Objective

To Identify the right execution channel for each customer

Analyze customer profitability, transactions performed across channels, cost of channels, etc.

Develop channel optimization model

Overlay model with capacity available on the channels / availability of updated contact details of customer

Increase spend per card – Which is the best channel?

Channel usage analysis at a customer level (# and types of transactions at each channel)

- Branch
- ATM
- Phone Banking
- Online Banking

\+

Calculate customer profitability (can use revenue as a proxy)

Customer ID	Customer Profitability	Segment
A1	448	High
B1	337	Medium
C1	219	High

\+

Map channels to customer

Customer	B1
Channels Used	Call Centre & Internet
Current Profitability	Medium
Channels preferred (based on What to Offer)	SMS, Online, Call centre

Offer campaign to Customer B1 through Internet banking channel

Channel Identification & Optimization

helps maximize response rate by identifying right execution channel for a campaign (2/2)

Largest banks in the world, use our AI/ML solutions to drive organic growth

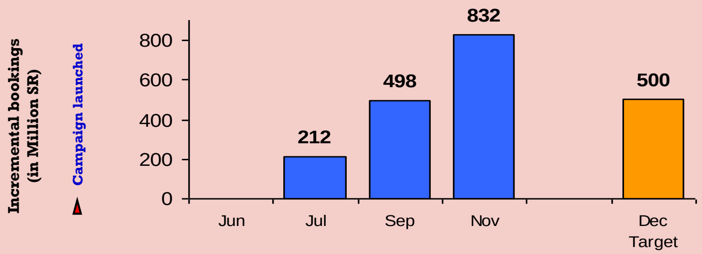
A robust implementation plan ensured flawless execution

Built a best-in-class analytics platform to maximize potential from the largely untapped customer base

Set-up infrastructure (people, process, data & culture) for AI/ML

Launch campaign based on AI/ML

Develop tracking & reporting mechanism to ensure timely review of ongoing campaigns



Beat Client Target by over 50%

- Capacity to Launch **25 campaigns end-to-end** based on AI/ML models for complete customer lifecycle management
- Developed SMS, email, and Call centre scripts for the campaign & **identified optimal channel for each customer in target base**
- Prepared campaign tracking and reporting dashboard **for tracking at all levels**
- **Trained & monitored performance** of sales agents for best results of the campaign
- Scope now extended to customer management & automation of CRM process to **enable full utilisation of all channels for customer management**

Booked an incremental revenue of USD **80** million in **2** years

Reduced cost of acquisition by **30%**

DIFFERENTIATED MODELLING APPROACH FOR MARKET MIX MODELLING FOR A LEADING CPG

Business Objective

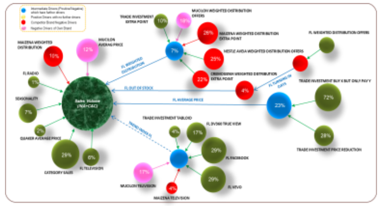
Develop a business oriented & statistically robust modeling approach to perform Market Mix Modelling (MMM) for Nestle's cereal brand. Generate insights to determine optimal level of investment across media channels

Modelling Approach

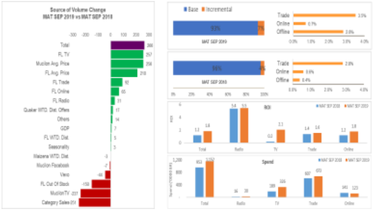
- Exploratory Data Analysis
- SEM / Driver Analysis
- Linear Model
- Bayesian Model (Markov Chain Monte Carlo Simulation)
- Optimizer & Simulator

Key Deliverables

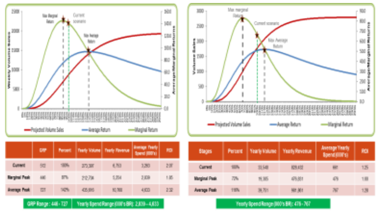
Identified direct and indirect drivers impacting the brand volume sales



Identified direct and indirect drivers impacting the brand volume sales



Determined optimum investment levels for different media interventions



Opportunities/Benefits Delivered

2Mn revenue opportunity identified through budget reallocation

Benefits

Identified that decreasing *Facebook investment* by 60K using the same in *Google Search* would give additional volume of 0.5% & additional revenue of 1.2M

Identified that reducing *Trade promotion* "Buy X Pay Y" investment by 420K and utilising it for Trade promotion "*Price Reduction*" would give additional volume of 0.2% and additional revenue of 600K

Identified that investment of extra 50K in Radio would give additional volume of 0.1% and additional revenue of 292K

LEADING BEVERAGE MANUFACTURER TO OPTIMALLY ALLOCATE BUDGETS AND THEREBY MAXIMIZE MARKETING ROI

Business Objectives

- Understand the Return on Investment of various marketing activities for one of their premium beverage brands(Brand1)
- What was the impact of Brand1's marketing programs on intermediate outcomes and consumption(household)?

Solution Approach

Exploratory data Analysis

Min, Max, Mean, Sum, Counts, Correlations, Trends, Seasons and Environmental Effects

SEM / Drivers Analysis

To understand the interactions, synergies, path and nature of relationships

Linear Mixed Model

Quantify impact of input variables on household consumption and intermediate variables (used the models as priors for Bayesian model).

Bayesian Model

Optimize the quantification of impact of different marketing programs on household consumption and intermediate variables

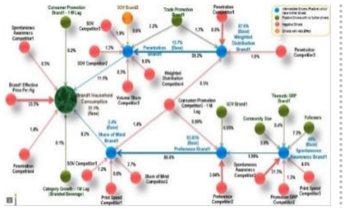
Final Insight & Simulator

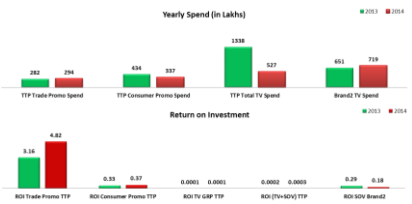
Choose the best model that makes business sense and is statistically valid for computing contributions & RoI's and to build Scenario Simulator

Analysis Output

Key insights emerged:

- Competitor brands' consumer promotional activities posed considerable threat to Brand1
- Brand1 thematic TV advertisement had long-term impact on spontaneous awareness of Brand1
  - The build-up time was around 6 months.
- Umbrella campaigns had a significant impact on share of mind





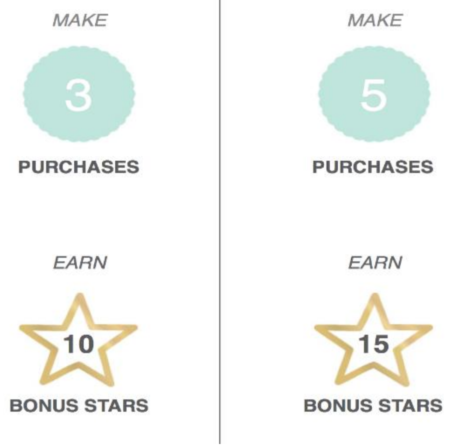
Business Impact

Enabled 8% savings through reallocation of TV budgets to alternate channels which command a higher ROI

10% saving in TVC spends

STAR DASH – A QUICKER WAY TO EARN BONUS

A MAJOR CAMPAIGN RUN BY SBUX TO INCREASE ENGAGEMENT

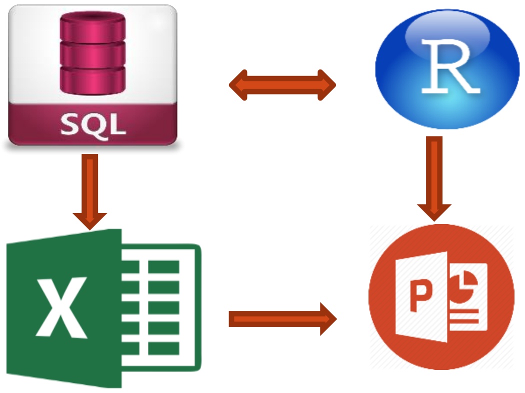


Business Needs

- **Design** a game-like personalized offer to increase the engagement of its loyalty members by giving them a target to achieve and win additional reward points as an incentive.
- Follow-up with a **PCA** (post campaign analysis) to understand the **impact** of the campaign

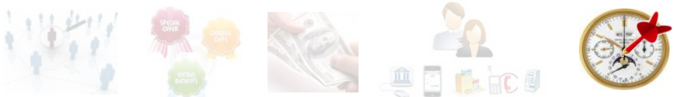
Solution

- **Design** the campaign according to the business objective and the technological & data limitations, with cross-functional teams
- **Execute** the program through identifying key member segments, designing their targets, tracking their behavior and awarding points



Estimated Benefits

- Determination of Incremental and campaign efficiency
- Identification of customer segments outperforming other segments
- A full-fledged study of the frequency behavior of loyalty members



5 WHEN TO TARGET – HELPS GAIN SHARE OF TIME BY RIGHT-TIMING THE OFFER IN NEAR REAL-TIME

Marketing campaigns launched based on events/triggers receive the highest response from customers, as qualified by Research

Objective: Identify the events significant to customers (Lifestage changes, significant change in transaction pattern, location, etc.)

Analyze data to build a library of relevant events and track these in a real-time basis – Triggering of rules will provide clues for timing of campaigns

Map the events to a library of campaigns

Event based marketing (EBM) is typically 5-10 times more effective than traditional marketing efforts (based on end-of-month data)

Illustrative

Increase spend per card – When to target?

Develop a library of events → …Monitor switch data in near real-time to check if rules have been triggered → …as soon as event is identified, send campaign to customer using right channel

Examples:

1. Customer ABC is identified as Medium Value, Retail only spend; Nationality is Turkey; Uses Internet Banking; From Market basket analysis - Retail and airlines go together; Hence, when customer logs onto Internet Banking channel on a weekend, offer a promotion on Singapore Airlines to his hometown

2. Customer XYZ is identified as Low Value, Likely to move to Medium Value, has used her card at Japanese restaurants in past, etc. – If she transacts at ATM xyz between 1130 and 1 pm on friday, then send her an offer for Lunch at Japanese restaurants within 0.5 km of ATM

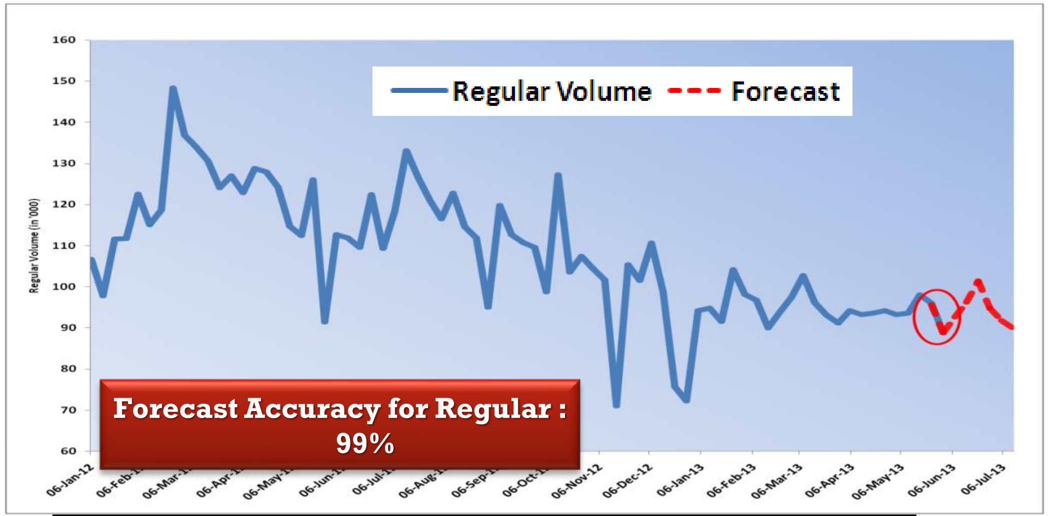
Incorporate advance analytics to forecast incoming claims volume

1 **Objective: Improve operational efficiency of claims processing by forecasting the volume of incoming claims**

- Client wanted to have an in-depth analysis of incoming volume for three LOB's – Regular, SMAP & CDC.
- Based on the above, forecast the incoming claims volume at weekly level for 2 months

2 **Approach: Time series model is used to forecast the incoming claim volume**

- **Data study / Treatment-**
  - ✓ In-depth data mining over a period of 3.5 years to understand data behavior
  - ✓ Data validation and treatment by studying the trends and seasonality in data
  - ✓ Removal of outlier values to improve the accuracy of model.
- **Data modeling steps-**
  - ✓ Check for stationarity and seasonality in data
  - ✓ Testing different models –VARMAX & UCM for forecasting the claims volume.
  - ✓ Selection of model based on specific statistics and validation parameters



Regular Volume Forecast				
Week ending on	Regular Volume	Forecast	Deviations (Error %)	Accuracy
5/10/2013	95,990	95,626	-0.4%	98.45%
5/17/2013	88,817	88,787	0.0%	98.43%
5/24/2013		92,628		
5/31/2013		96,241		
6/7/2013		101,439		
6/14/2013		94,791		
6/21/2013		91,803		
6/28/2013		89,854		

Productivity Improvement:
2 FTE's per Month
YoY Gain – $48,000

Advanced Forecasting Modeling Techniques considered and used

Low — Complexity — High

ARIMA

- Time Series with trend
- Captures trend and pattern over time
- Series need to be stationary to finalize model
- Can be extended for seasonal data using SARIMA model
- Powerful for short term forecasting

UCM (Unobserved Component Model)

- Captures trend, seasonal, and cyclical effects
- Use Hodrick-Prescott (HP) filter concepts decomposition and analyzing the series more correctly
- Captures very high or low fluctuations
- Consider the effect of other input series

Best Short Term Prediction

BSTS (Bayesian Structural Time Series) – ML Technique

- Time Series with feature selection
- More transparent because its representation does not rely on differencing, lags and moving averages
- Handles uncertainty in a better way because you can quantify the posterior uncertainty

MAPA (Multiple Aggregation Prediction Algorithm) – ML Technique

- Uses temporal aggregation to improve upon the established exponential smoothing family of methods
- Using multiple temporal aggregation gives robust and reliable automatic models
- Good performance and robust automatic specification

LSTM (Long Short Term Memory) – ML Technique

- Used to handle long/short run forecast
- Works on memory blocks that are connected through layers
- Complex computation been performed in each block
- LSTM works better with large size of data
- Used to make prediction for longer time period

Prophet – ML Technique

- Complex execution, but can capture effects of non linearity
- Interpretation of the results also requires expertise
- Captures Holiday Effect along with Trend and seasonality
- Robust to missing data, outliers and any dramatic changes in time series

Best Short & Long Term Prediction

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Based On Business Objectives, We Analyze Customer Data To Develop Analytical Models Across The Customer Life Cycle To Assist Marketing Initiatives

Thank You